

Lecture 11: Inequalities II

Recall the geometric mean and arithmetic mean inequality for two numbers

$$\sqrt{ab} \leq \frac{a+b}{2}$$

and the equality holds only if $a = b$.

Claim 1: if the sum of two numbers are the same, the maximal product value is attained by they are the same. This is equivalent to square gives the maximal area given the fixed perimeter.

Claim 2: if the product of two numbers are the same, the minimal sum value is attained by they are the same. This is equivalent to square gives the minimal perimeter given the fixed area.

Problem: Find maximal ab if $a + 3b = 10$.

Solution: By GM-AM inequality, we have

$$\sqrt{3ab} \leq 5,$$

with equality holds when $a = 3b$. So the maximal value of ab is $25/3$.

GM-AM inequality for three numbers

$$\sqrt[3]{abc} \leq \frac{a+b+c}{3}$$

The proof is given in last lecture using the factorization $a^3 + b^3 + c^3 - 3abc$.

Lemma: equality holds if and only if $a = b = c$.

Proof: if $a \neq b$, then by Claim 1

$$ab < \left(\frac{a+b}{2}\right)^2$$

So we have

$$\sqrt[3]{abc} < \sqrt[3]{\left(\frac{a+b}{2}\right)^2 c} \leq \frac{\frac{a+b}{2} + \frac{a+b}{2} + c}{3} = \frac{a+b+c}{3}.$$

So $a = b$ and by the same reason $a = c$. So $a = b = c$.

The proof works for the general GM-AM inequality for n numbers.

Theorem: GM-AM inequality for n nonnegative numbers

$$\sqrt[n]{a_1 a_2 \cdots a_n} \leq \frac{a_1 + a_2 + \cdots + a_n}{n}$$

where the equality holds only if $a_1 = a_2 = \cdots = a_n$.

Proof when $n = 4$: We use a, b, c, d instead. Notice that

$$ab \leq \left(\frac{a+b}{2}\right)^2$$

So we have

$$abcd \leq \left(\frac{a+b}{2}\right)^2 \left(\frac{c+d}{2}\right)^2 \leq \left(\frac{a+b+c+d}{4}\right)^4.$$

This works for $n = 2^k$.

Proof when $n = 3$ using $n = 4$: Consider $a, b, c, \sqrt[3]{abc}$. We have

$$\sqrt[4]{abc \sqrt[3]{abc}} \leq \frac{a + b + c + \sqrt[3]{abc}}{4}$$

$$\sqrt[4]{abc^{\frac{4}{3}}} \leq \frac{a + b + c + \sqrt[3]{abc}}{4}$$

$$\sqrt[3]{abc} \leq \frac{a + b + c + \sqrt[3]{abc}}{4}$$

$$4\sqrt[3]{abc} - \sqrt[3]{abc} \leq a + b + c$$

$$\sqrt[3]{abc} \leq \frac{a + b + c}{3}.$$

This works when $n = 8$ then apply the same trick to $n = 5, 6, 7$.

Problem 331 Prove

$$\left(1 + \frac{1}{n}\right)^n < \left(1 + \frac{1}{n+1}\right)^{n+1}$$

Solution

$$\sqrt[n+1]{\left(1 + \frac{1}{n}\right)^n} = \sqrt[n+1]{1 \cdot \left(1 + \frac{1}{n}\right)^n} \leq \frac{1 + n\left(1 + \frac{1}{n}\right)}{n+1} = 1 + \frac{1}{n+1}$$

Problem Prove

$$\left(1 + \frac{1}{n}\right)^n < 3$$

Solution: By binomial expansion, we have

$$\begin{aligned}\left(1 + \frac{1}{n}\right)^n &= 1 + \binom{n}{1} \frac{1}{n} + \binom{n}{2} \frac{1}{n^2} + \cdots + \binom{n}{n} \frac{1}{n^n} \\&= 1 + \frac{n}{1} \frac{1}{n} + \frac{n(n-1)}{2!} \frac{1}{n^2} + \frac{n(n-1)(n-2)}{3!} \frac{1}{n^3} \cdots + \binom{n}{n} \frac{1}{n^n} \\&< 1 + 1 + \frac{1}{2!} + \frac{1}{3!} + \cdots + \frac{1}{n!} \\&< 1 + 1 + \frac{1}{1 \cdot 2} + \frac{1}{2 \cdot 3} + \cdots + \frac{1}{(n-1)n} = 3 - \frac{1}{n} < 3\end{aligned}$$

Theorem:

$$e = \lim_{n \rightarrow \infty} \left(1 + \frac{1}{n}\right)^n = 1 + 1 + \frac{1}{2!} + \frac{1}{3!} + \cdots = 2.718281828459045\ldots$$